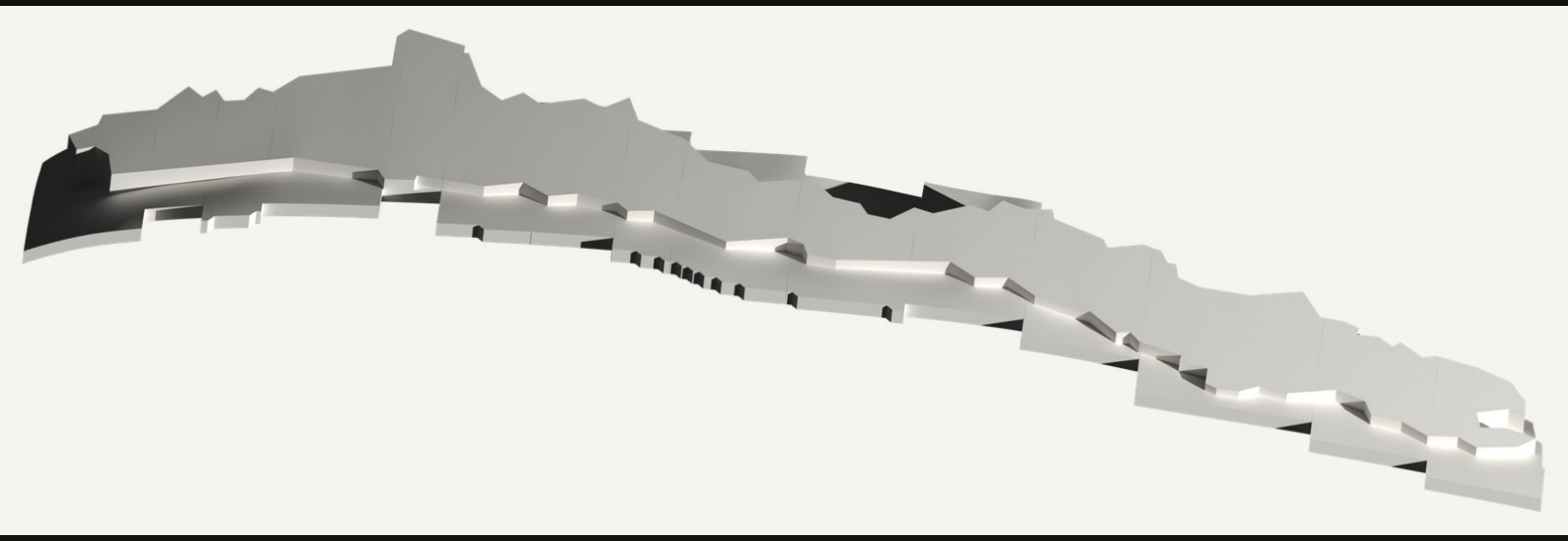
DOSSIER OF THE WORK · OCTOBER 2026



El sonido del movimiento.

[The sound of movement] An exhibition of seismology, music and heritage

Bastián I. Garrido Kogan

Author and director · Civil engineer, MSc in earthquake engineering

bgarridosys.github.io/sonido-del-movimiento

SUMMARY

A country turned into an instrument.

El sonido del movimiento [The sound of movement] is an immersive exhibition that presents seismology as part of Chile's cultural heritage. In a single room, visitors assemble the **relator** by hand: an instrument of 26 3D-printed sections that reproduces Chile from 17°S to 43°S over the subduction of the Nazca plate. A mapping is projected onto it and, on four panels, an animation and the transcript of the story. In nine minutes, six moments travel through five hundred years of earthquakes: from the Mapuche story of Tren Tren Vilu and Caicai Vilu to colonial chronicles, the birth of modern seismology with Raúl Madariaga and José Vicente Asuar, the seismic records of 11 September 1973 and the seismic silence of the pandemic. The music travels backwards in time, from artificial intelligence to natural sounds. The work brings together science, art and memory, and offers a printable kit so that schools can build it.

26
sections of one degree of latitude each

17–43°S
mainland Chile 3D-printed

6
moments, from belief to science

9:07
minutes of visit in a single room



Audiovisual mock-up V5 of the room: the relator at the centre and the panels on the walls (moment M2).

Project status · October 2026

Component	Status
Relator	Printed at full size (26 sections, 3.29 m); presented on 16 May 2026.
Support	Completed on 28 August 2026; closes the construction process.
Script and audio	Complete for the six moments (9 min 07 s).
Audiovisual mock-up	Version V5 of the whole room, with a 360° visit for virtual-reality headsets.
Panel animation	Working sketches made with AI assistance; the final animation will be drawn by Ana Norambuena.
Installation	Technical sheet defined; venue to be confirmed.

THE WORK

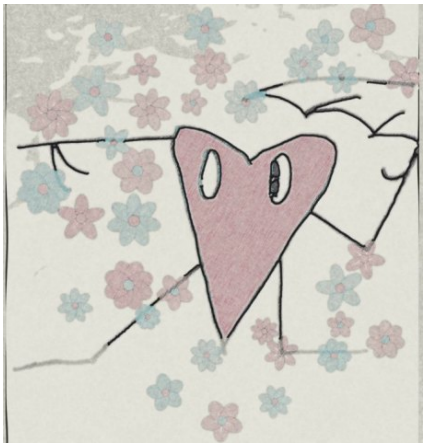
Four panels. One relator.

Two pairs of panels surround the relator: those on the east and west walls show a cartoon-style animation, and those on the north and south walls, the transcript of the story. A mapping telling the same story is projected onto the relator. The moments advance chronologically with the progress of seismology, supported by excerpts from scientific papers.

The audience is part of the work: in moment 0 visitors assemble the relator with their own hands, section by section, and then surround it on both sides to listen.

Six moments

	Moment	Period	Heritage	Content
M0	**The creation**	Today	Community	The audience assembles the country, section by section.
M1	**Ancestral Chile**	Before 1500	Intangible	The Mapuche story of Tren Tren Vilu and Caicai Vilu: the transformation of the land.
M2	**From the Colony to the Republic**	1500–1970	Documentary	Colonial chronicles and the 1730 tsunami, also recorded in Japan; “Epopeya de Chillán” (1939).
M3	**Turning point**	1970s	Scientific and technological	Madariaga, Asuar and the COMDASUAR: the beginning of modern seismology in Chile.
M4	**Memories I**	11 Sep. 1973	Memory	The seismic records of 11 September 1973.
M5	**Memories II**	2020	Natural	The seismic silence of the pandemic and the return of the forest.



M2: the heart surrounded by flowers that regains its colour (AI-assisted sketch).



M3 panel: Madariaga and Asuar, fictional recreation (AI-assisted sketch).

SEISMOLOGY AS HERITAGE

Movement defines us too.

We tend to think of heritage as something still: palaces, stone monuments, folk traditions. But heritage is also what shapes our social practices, passes values from one generation to the next and takes on new meaning over time. Living in Chile means living with the heartbeat of the Earth. The memory recorded in a seismogram, the oral accounts of those who survived great disasters and the growth of the scientific knowledge that protects us form an intangible and natural legacy.

> *"I want to show you how the movement of the Earth defines us as people, and our surroundings, from belief to scientific knowledge, and how we too can affect it and see it through the memory of a seismogram."*
>
> — Bastián I. Garrido Kogan

AUDIENCE AND ACCESSIBILITY

Designed for children, families and schools.

Scale	The top of the relator is at 0.80 m: a 1.2 m child or a wheelchair user can see it from end to end and help assemble it.
Participation	The audience builds the work in M0; the experience is tactile, sonic and visual.
Heritage education	A printable kit in three scales to build the relator at school, with a guide for each arch.
Languages	Website in Spanish, English and Japanese; transcript of the story on the panels and narration subtitles in the web videos.
Remote	Video tour from three viewpoints and a 360° visit for virtual-reality headsets.



The full-size relator, with the numbered scale model in the foreground.



The full-size relator, held upright by the author.

TEAM

Who makes the work.

Name	Role	Contribution
Bastián I. Garrido Kogan	Author and director	Concept, script, 3D modelling of the relator, installation design and direction. Civil engineer, MSc in earthquake engineering.
Ana Norambuena	Illustration and animation	Author of the cartoon “Origen tectónico” and of the panels’ style; she will draw the final animation.
Fabián Astudillo	3D printing	Printed the 26 scale sections (August 2025).
Enrique Neira	Script	Structure of the script by moments and the name “relator”.
Diego Valenzuela and Carlos Benavides	Musical advice	Musical background for the soundtrack.
Cristian Bastías Curivil	Advice · M1	Geologist and researcher of the Mapuche worldview; conversation on the story of Tren Tren Vilu and Caicai Vilu (January 2026).

References and protagonists: Raúl Madariaga, José Vicente Asuar, Sergio Ruiz, Rodolfo Saragoni, Violeta Parra, Nicanor Parra and Jorge Peña Hen.

PROCESS

From paper to the room.

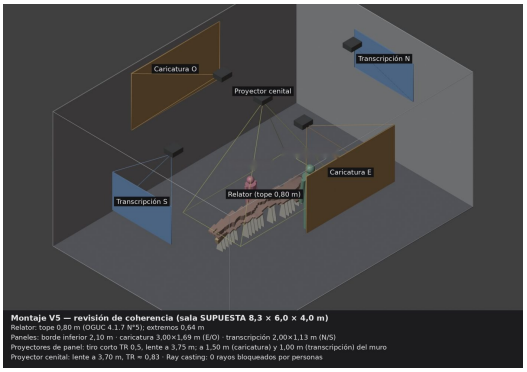
2022–2024	Origin in teaching: paper models of subduction in courses at the Universidad de Chile.
Jan. 2025	First piece printed at the FCFM FabLab, Universidad de Chile.
8 Aug. 2025	Fabián Astudillo prints the 26 scale sections; the idea of a musical instrument emerges.
2025	Script by moments with Enrique Neira; the name “relator” is born.
24 Sep. 2025	Visit to “Herencia sísmica”, SENAPRED’s exhibition room.
9 Dec. 2025	Visit to “Cómo armar una máquina en 10 pasos”, by M. Ignacia C. X. Valdebenito G.: inspiration for M0.
6 Jan. 2026	Conversation with Cristian Bastías Curivil on the Mapuche story (M1).
Jan.–May 2026	Printing of the full-size relator (3.29 m).
16 May 2026	Progress presentation.
28 Aug. 2026	Final support; end of the construction process.
2026	Digital installation in Blender, audiovisual mock-up V5 and 360° visit.

TECHNICAL SHEET

What the room needs.

Reference room	8.3 × 6.0 m, height 4.0 m; darkened (≤ 5 lux during projections).
Relator	26 3D-printed sections, 17°S–43°S; top at 0.80 m; on a support of 5 arches.
Projection	1 overhead laser projector ≥ 4,000 lm on the relator; 4 short-throw projectors ≥ 3,000 lm for the panels.
Screens	2 cartoon screens of 3.00 × 1.69 m (east and west walls) and 2 transcript screens of 2.00 × 1.13 m (north and south).
Playback	Computer with a GPU with ≥ 3 outputs, timeline software (e.g. QLab or Millumin), backup machine.
Audio	4-output interface, 4 active speakers in the corners and optional subwoofer; 75–85 dB(A).
Power	2 circuits of 16 A / 220 V; estimated load ≈ 3 kW; UPS for the computer.
Duration	9 min 07 s per show; M0 adapts to the pace of assembly.

Full detail by category in the “Technical sheet” section of the website.



Digital installation of the room in Blender (model V5).

RIGHTS AND USE OF ARTIFICIAL INTELLIGENCE

Own work	Texts, photographs, models and animations of his own are by Bastián I. Garrido Kogan. Third-party works are cited with their source.
Web edition	The public videos omit protected works (Parra, Chopin/Asuar, Peña Hen and the synthetic voice of M1); the full audio is reserved for the room, under the corresponding licences.
Educational kit	The printing kit files are released under a Creative Commons BY-NC 4.0 licence: free use and adaptation with attribution, for non-commercial purposes.
Artificial intelligence	The audiovisual mock-up, the animation sketches and the composition were made with AI assistance (Claude, by Anthropic), from the author’s script, sketches and audio. The panel sketches follow Ana Norambuena’s style as a guide; the final animation will be drawn by her. The images in “First visualizations” were generated with Google Gemini. The meeting of Madariaga and Asuar in M3 is fictional.

LINKS

Website	bgarridosys.github.io/sonido-del-movimiento
360° visit (YouTube)	youtu.be/VFVwytNrvDk